

Experiment: INNER/OUTER JOINS

Environment: Microsoft Windows

Tools/ Language: Oracle/SQL

Objective: Write the SQL queries using Set Operations and Joins.

Theory & Concepts:

SQL JOINS are used to retrieve data from multiple tables. A SQL JOIN is performed whenever two or more tables are joined in a SQL statement.

There are different types of SQL joins:

SQL INNER JOIN (or sometimes called simple join)

SQL CROSS JOIN

SQL NATURAL JOIN

SQL LEFT OUTER JOIN (or sometimes called LEFT JOIN)

SQL RIGHT OUTER JOIN (or sometimes called RIGHT JOIN)

SQL FULL OUTER JOIN (or sometimes called FULL JOIN)

SQL INNER JOIN (SIMPLE JOIN)

SQL INNER JOINS return all rows from multiple tables where the join condition is met.

Syntax

The syntax for the SQL INNER JOIN is:

SELECT columns

FROM table1

INNER JOIN table2

ON table1.column = table2.column;

If the tables COUNTRIES and CITIES have two common columns named POPULATION and COUNTRY_ISO_CODE, JOIN applies equality condition on ISO codes with cities having less POPULATION attributes:

```
SELECT * FROM
COUNTRIES
INNER JOIN CITIES
On COUNTRIES. COUNTRY_ISO_CODE=CITIES. COUNTRY_ISO_CODE
And COUNTRIES.POPULATION > CITIES.POPULATION;
```

SQL LEFT OUTER JOIN

Another type of join is called a LEFT OUTER JOIN. This type of join returns all rows from the LEFT-hand table specified in the ON condition and only those rows from the other table where the joined fields are equal (join condition is met).

Syntax

The syntax for the SQL LEFT OUTER JOIN is:

SELECT columns

FROM table1

LEFT [OUTER] JOIN table2

ON table1.column = table2.column;

In some databases, the LEFT OUTER JOIN keywords are replaced with LEFT JOIN.

```
SELECT * FROM
COUNTRIES
LEFT JOIN CITIES
On COUNTRIES. COUNTRY_ISO_CODE=CITIES. COUNTRY_ISO_CODE
And COUNTRIES.POPULATION > CITIES.POPULATION;
```

SQL RIGHT OUTER JOIN

Another type of join is called a SQL RIGHT OUTER JOIN. This type of join returns all rows from the RIGHT-hand table specified in the ON condition and only those rows from the other table where the joined fields are equal (join condition is met).

Syntax

The syntax for the SQL RIGHT OUTER JOIN is:

SELECT columns

FROM table1

RIGHT [OUTER] JOIN table2

ON table1.column = table2.column;

In some databases, the RIGHT OUTER JOIN keywords are replaced with RIGHT JOIN.

```
SELECT * FROM
COUNTRIES
RIGHT JOIN CITIES
On COUNTRIES. COUNTRY_ISO_CODE=CITIES. COUNTRY_ISO_CODE
And COUNTRIES.POPULATION > CITIES.POPULATION;
```

SQL FULL OUTER JOIN

Another type of join is called a SQL FULL OUTER JOIN. This type of join returns all rows from the LEFT-hand table and RIGHT-hand table with nulls in place where the join condition is not met.

Syntax

The syntax for the SQL FULL OUTER JOIN is:

SELECT columns

FROM table1

FULL [OUTER] JOIN table2

ON table1.column = table2.column;

In some databases, the FULL OUTER JOIN keywords are replaced with FULL JOIN.

```
SELECT * FROM
COUNTRIES
FULL JOIN CITIES
On COUNTRIES. COUNTRY_ISO_CODE=CITIES. COUNTRY_ISO_CODE
And COUNTRIES.POPULATION > CITIES.POPULATION;
```

SQL NATURAL JOIN

A NATURAL JOIN is a JOIN operation that creates an implicit join clause for you based on the common columns in the two tables being joined. Common columns are columns that have the same name in both tables.

If the SELECT statement in which the NATURAL JOIN operation appears has an asterisk (*) in the select list, the asterisk will be expanded to the following list of columns (in this order):

- All the common columns
- Every column in the first (left) table that is not a common column
- Every column in the second (right) table that is not a common column

An asterisk qualified by a table name (for example, COUNTRIES.*) will be expanded to every column of that table that is not a common column.

Syntax

```
Select *  
FROM table1  
NATURAL JOIN table2;
```

Examples

If the tables COUNTRIES and CITIES have two common columns named COUNTRY and COUNTRY_ISO_CODE, NATURAL JOIN applies equality condition on both attributes:

```
SELECT * FROM COUNTRIES NATURAL JOIN CITIES;
```

CROSS JOIN operation

A CROSS JOIN is a JOIN operation that produces the Cartesian product of two tables. Unlike other JOIN operators, it does not let you specify a join clause. You may, however, specify a WHERE clause in the SELECT statement.

Examples

The following SELECT statements are equivalent:

```
SELECT * FROM CITIES CROSS JOIN  
FLIGHTS
```

```
SELECT * FROM CITIES, FLIGHTS
```

Practical Assignment

```
drop table College;
drop table Student;
drop table Apply;

create table College(collegeName varchar2(10) primary key, state
varchar2(10), enrollment int);
create table Student(sID int primary key, sName varchar2(10), GPA
real, sizeHS int);
create table Apply(sID int, cName varchar2(10), major varchar2(20),
decision char(1), primary key(sID, major, cName), constraint sID_fk
Foreign key(sID) references Student, constraint cName_fk Foreign
key(cName) references College);

delete from Student;
delete from College;
delete from Apply;

insert into Student values (123, 'Amy', 3.9, 1000);
insert into Student values (234, 'Bob', 3.6, 1500);
insert into Student values (345, 'Craig', 3.5, 500);
insert into Student values (456, 'Doris', 3.9, 1000);
insert into Student values (567, 'Edward', 2.9, 2000);
insert into Student values (678, 'Fay', 3.8, 200);
insert into Student values (789, 'Gary', 3.4, 800);
insert into Student values (987, 'Helen', 3.7, 800);
insert into Student values (876, 'Irene', 3.9, 400);
insert into Student values (765, 'Jay', 2.9, 1500);
insert into Student values (654, 'Amy', 3.9, 1000);
insert into Student values (543, 'Craig', 3.4, 2000);
insert into College values ('Stanford', 'CA', 15000);
insert into College values ('Berkeley', 'CA', 36000);
insert into College values ('MIT', 'MA', 10000);
insert into College values ('Cornell', 'NY', 21000);
insert into College values ('Harvard', 'MA', 50040);
insert into Apply values (123, 'Stanford', 'CS', 'Y');
insert into Apply values (123, 'Stanford', 'EE', 'N');
insert into Apply values (123, 'Berkeley', 'CS', 'Y');
insert into Apply values (123, 'Cornell', 'EE', 'Y');
insert into Apply values (234, 'Berkeley', 'biology', 'N');
insert into Apply values (345, 'MIT', 'bioengineering', 'Y');
insert into Apply values (345, 'Cornell', 'bioengineering', 'N');
insert into Apply values (345, 'Cornell', 'CS', 'Y');
insert into Apply values (345, 'Cornell', 'EE', 'N');
insert into Apply values (678, 'Stanford', 'history', 'Y');
insert into Apply values (987, 'Stanford', 'CS', 'Y');
insert into Apply values (987, 'Berkeley', 'CS', 'Y');
insert into Apply values (876, 'Stanford', 'CS', 'N');
insert into Apply values (876, 'MIT', 'biology', 'Y');
insert into Apply values (876, 'MIT', 'marine biology', 'N');
insert into Apply values (765, 'Stanford', 'history', 'Y');
insert into Apply values (765, 'Cornell', 'history', 'N');
insert into Apply values (765, 'Cornell', 'psychology', 'Y');
insert into Apply values (543, 'MIT', 'CS', 'N');
```

Practice Exercise:-

Student

sID	sName	GPA	sizeHS
123	Amy	3.9	1000
234	Bob	3.6	1500
345	Craig	3.5	500
456	Doris	3.9	1000
567	Edward	2.9	2000
678	Fay	3.8	200
789	Gary	3.4	800
987	Helen	3.7	800
876	Irene	3.9	400
765	Jay	2.9	1500
654	Amy	3.9	1000
543	Craig	3.4	2000

College

collegeName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

Write SQL Queries for the following:

- Q1.** Produce a combine table in which each student is combine with every other application.
- Q2.** Give Student ID, name, GPA and name of college and major each student applied to.
- Q3.** Find detail of applications who applied to California State.
- Q4.** IDs, name, GPA of students and name of college with GPA > 3.7 applying to Stanford
- Q5.** Find detail of Student who apply to CS major and their application are rejected
- Q6.** Find detail of student and application who applied to colleges at New York
- Q7.** Find detail of student who have not applied to any of college
- Q8.** Find college where no student have applied
- Q9.** Find sID who have only one application
- Q10.** Find name and GPA of applicants who apply to any college whose enrollment is not more than 25000.
- Q11.** Find pair of students (sID) having same GPA. (each pair should occur just once in result)

Exercise

For each of the following you need to write three queries

i.e. three version first using :CROSS Join

Second using: Natural Join

And third using: Inner Join

You are also advised to observe output of all three

Q12. Find student and major he / she applied to.

Q13. Find detail of student who came from high school have size less than 20000 and applied to CS at Stanford.

Q14. Provide complete detail of each student where they applied what major they applied to what was the decision and complete detail of college they applied.

Q15. Names and GPAs of students with HS>1000 who applied to CS and were rejected

Q16. Names and GPAs of students with HS>1000 who applied to CS at college with enr>20,000 and were rejected

Pre Experiment Questions

1. When we need to combine two tables?
2. Difference between Equi Join and Theta Join
3. Difference between Natural join and Inner Join

Post Experiment Questions

1. When can we use natural join?
2. When we are bound to use inner join?
3. Can we implement all joins using cross join?
4. Where and in what kind of queries require outer joins?